Coding Practice Set - 04

1-D Array Logic & Searching

1. **Write a program to calculate the sum of numbers stored in an array of size 10. Take array values from the user.**
2. **Write a program to calculate the average of numbers stored in an array of size 10. Take array values from the user.**
3. **Write a program to calculate the sum of all even numbers and the sum of all odd numbers, which are stored in an array of size 10.**
4. **Write a program to find the greatest number stored in an array of size 10.**
5. **Write a program to find the smallest number stored in an array of size 10.**
6. **Write a program to search for a specific target value in an array of size 10 using Linear Search.**

Modular Array Functions (User-Defined)

7. **Write a function to find the greatest number from a given array of any size using the Takes Something Returns Something (TSRS) approach.**
8. **Write a function to find the smallest number from a given array of any size using the TSRS approach.**

9. Write a function to sort an array of any size in ascending order using the TSRS approach.
10. Write a function to reverse the elements of an array of any size in place.

2-D Grid & Matrix Mathematics

11. Write a program that asks the user for the number of rows R and columns C, inputs the elements for a matrix of that size, and prints it in a clean grid format.
12. Write a program to find and print the sum of each individual row in a 3 * 3 matrix.
13. Write a program to calculate and print the sum of each individual column in a 3 * 4 matrix.
14. Write a program that searches a 3 * 3 matrix and counts how many times a user-specified number K appears inside the grid.
15. Write a program that scans a 3 * 3 matrix, replaces all negative numbers with a 0, and then displays the updated matrix.
16. Write a program to scan a 3 * 3 matrix and print the largest element present in each horizontal row.
17. Write a program to display only the elements that fall on the main diagonal (from top-left to bottom-right) of a 3 * 3 matrix, setting all other positions to zeros.
18. Write a program to calculate the total combined sum of both the main diagonal (top-left to bottom-right) and the secondary diagonal (top-right to bottom-left) in a 3 * 3 matrix.

19. **Write a program that inputs a 4 * 4 matrix and prints only its outer boundary elements, leaving the inner core elements blank.**
20. **Write a program that multiplies every single element of a 3 * 3 matrix by a constant integer value entered by the user.**